Drilling drawworks fail-safe system disc brakes

Health and Safety Executive - Safety notice

Department name: Energy Division (ED)

Bulletin number: ED02-2026 Issue date: 09/26 Target audience: Offshore installation operators and duty holders, drilling plant and equipment manufacturers and suppliers

Issue

There is a risk of an uncontrolled descent of top drives for certain electrically motor-driven drawworks with fail-safe functionality provided by disc brakes. There may be a danger to persons when disc brakes are used to stop dynamic loads in an emergency as they may not function as intended despite successful static brake capacity testing.

Outline of the problem

Certain electrically motor-driven drilling drawworks utilise disc brakes as a fail-safe means of stopping and holding the travelling system during emergency conditions. HSE has identified that, although some of these disc brakes have successfully passed routine static holding-capacity tests, they may not provide adequate braking performance when required to stop a moving load dynamically. Failure of the disc brakes to arrest a descending load during an emergency stop condition could result in uncontrolled load descent, creating a significant risk of injury to personnel and damage to equipment.

Under normal operating conditions, load control is primarily provided by the electrically driven motors, while the disc brakes principally function as a parking or static holding brake. However, the disc brakes are also required to perform as an emergency fail-safe braking system when abnormal operating conditions occur.

An HSE investigation identified an event during drawworks operations when electrically driven motors were no longer able to provide load control, resulting in a dynamic load that required emergency stopping using the disc brakes. Although previous testing had confirmed that the brakes were capable of holding the required static load, they were unable to generate sufficient dynamic braking torque to arrest the descending load. This degradation in dynamic braking performance was not detectable through routine static brake testing and only became apparent when the brakes were required to perform their intended emergency dynamic braking function.

The investigation has identified some important findings.

Design requirement and initial suitability

• Industry specification and standards were not explicit in their requirements to dynamically test, verify and demonstrate the fail-safe emergency brake system functionality.

Testing and commissioning

• The testing and commissioning may not have been sufficient to provide an adequate demonstration of the fail-safe emergency brake system functionality and reliability.

Ongoing maintenance and performance testing

• Clear standards for maintenance and ongoing performance testing may not be fully in place or inadequate (for example, static capacity brake testing may be insufficient to confirm dynamic performance).

Other factors influencing dynamic braking performance

It’s worth noting that many other ongoing operational and maintenance factors may influence dynamic braking performance in an emergency. These may include, but are not limited to:

• contamination

• disc corrosion

• component wear, tolerance and alignment

• broken springs

• brake pad ageing and compression cycles

• ineffective or insufficient maintenance practices

• insufficient design assessments

• lack of original equipment manufacturer (OEM) supplied information and instruction

• thermal capacity of disc/pad system

• changes to supplied consumable parts

There may also be environmental design factors that influence the above, and these may present themselves differently on the same installation, for example enclosed or open drilling derricks and equipment.

Duty holders, OEMs and suppliers of Drawworks should consider the following actions:

Action required

Duty holders should:

• Review your case for safety, including checking what fail safe systems are in place for the Drawworks, to satisfy the requirements of the Provision and use of Work Equipment Regulations 1998, and the Lifting Operations Lifting Equipment Regulations 1998

• Review the FMEA or FMECA assessments provided by the OEM to establish what, if any, assessment has been undertaken for the dynamic performance of the emergency disc braking system (for example supplier ‘type approval’ testing or testing to support the declaration of conformity).

• Establish if any standards of performance have been detailed in these documents and if your maintenance management system contains these standards of performance.

• Ensure your planned maintenance routines for the Drawworks on your installation is as described in the instructions, and any product bulletins, provided for the equipment.

• Confirm with the OEM if all revisions to Drawworks instructions have been released via a product bulletin.

• Inspect the whole braking system for signs of damage, wear and corrosion, paying specific attention to the brake disc rotors for visible signs of corrosion and pitting, out of tolerance wear on hinged joints, cracked or broken springs within the brake cylinders, signs of pitting or other corrosion damage to springs and evidence of water/moisture ingress into the brake cylinders.

• Ensure brake callipers and pads are set to the critical tolerances as provided in the OEM instructions.

Equipment manufacturers and suppliers should:

• Review any design assessments, FMEA’s or FMECA’s still available for the Drawworks you manufacture and supply to establish how the dynamic performance of emergency disc brakes was considered (for example, ‘type approval’ method or other).

• Where information is no longer available, or you did not consider dynamic braking performance in your design assessments, undertake a design review of your product to assess the risks of poor dynamic braking performance and the inability to prevent a load from descending uncontrollably.

Your assessment must: Ensure disc brakes are dynamically rated for emergency use, taking account of the speed and the combined weight of the travelling block and string load in an emergency stop situation, that is, can you make a demonstration the system will fail safe within the stated operational parameters whilst considering the worst allowable conditions? (such as operating with broken springs or operating at maximum air gap due to component wear).

• Identify all degradation mechanisms that can develop within the disc brake system if the brakes are not dynamically activated for a period.

• Identify allowable wear tolerances for critical parts within the disc braking system.

• Evaluate the effects of harsh weather marine environments on the disc braking system. For example, if water ingress into internal parts of the brake system is possible, and the damage that can be initiated and caused without any warning being provided to Drawworks operators.

• The thermal effects on the system to establish how long it may take for friction fade to reduce the dynamic braking capacity.

• From your assessments, you must develop performance standards (if gaps exist) that can be safely implemented at site by the operators of your Drawworks. These standards of performance must be specific, measurable, achievable, relevant and timebound (SMART) to ensure the dynamic performance of the emergency braking system can be safely maintained at site.

• Update any information and instructions that were provided with the Drawworks you manufactured and / or supplied, and ensure that these updates are communicated to the owners of affected Drawworks.

Guidance

• HSE Technical guidance on the safe use of lifting equipment offshore HSG221 (Second edition)

• HSE offshore information sheet 14/2007 Guidance on regulations and testing applicable to drawworks equipment

• BS EN IEC 60812:2018 Failure modes and effects analysis (FMEA and FMECA) on the Bristish Standards Institution website

• API Specification 7K: Drilling and Well Servicing Equipment on the American Petroleum Institute website

Relevant legal documents

On legislation.gov.uk:

• section 2(1), section 3(1) and section 6 of the Health and Safety at Work etc Act 1974

• Provision and Use of Work Equipment Regulations 1998

• Lifting Operations Lifting Equipment Regulations 1998

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